



CIS 9660 PROJECT DRIVEN

a data mining project for accidents and safety

*towards
data driven
road safety*

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BACKGROUND AND INTRODUCTION

Road accidents are a significant public safety concern, causing loss of life, economic burdens, and infrastructural challenges. Understanding the patterns and factors behind road accidents can help policymakers, urban planners, and law enforcement agencies mitigate risks and develop safer road infrastructures for drivers, pedestrians, and the community. This project will analyse ten years of UK accident data to identify key factors contributing to road incidents and explore predictive models that could aid in accident prevention.

DATA SET AND DATA SOURCE

The [dataset](#) selected for this project, available on Kaggle, offers a decade-long (2005-2014) record of road accidents in the UK. The dataset comprises multiple key variables and data sets categorized into three main sections: Accident, Vehicle, and Casualty. These variables provide comprehensive insights into road accidents in the UK over a ten-year period. It provides a comprehensive overview, including geographical data, accident severity, environmental conditions, and vehicle-related details. This makes it an ideal dataset for analysing trends, identifying high-risk zones, and predicting accident severity using machine learning models.

RESEARCH QUESTION

We are interested in investigating two research questions which are **highly inferential** in nature and focus:

- 1. What are the contributing factors to accident severity? (Inference Problem)**
 - Our project aims to investigate the common characteristics and causes of accidents and the factors that lead to increased accident severity.
 - We examine how **environmental conditions** (light, weather, road), **driver characteristics** (gender and age), and **vehicle characteristics** (type, age) influence accident outcomes.
- 2. What are the factors leading to casualty severity? (Inference Problem)**
 - Our research aims to identify which groups are most vulnerable to severe outcomes in road accidents and factors that contribute to high casualty severity.
 - We examine how **casualty profile** (role, age, gender), **driver characteristics** (gender and age), and **vehicle characteristics** (type, age) influence casualty outcomes.

DATA DESCRIPTION

A total of 1,640,597 observations (rows) and 12 variables (columns) comprises the data set. In the first research question, the accident severity dependent variable itself has seven independent variables. Similarly the second research's casualty severity dependent variable also has seven independent variables.

Research Question 1: <i>What are the contributing factors to accident severity?</i>			Research Question 2: <i>What are the factors leading to casualty severity?</i>		
Dependent Variable			Dependent Variable		
Accident Severity	Ord	Fatal(1), Serious(2), Slight(3)	Casualty Severity	Ord	Fatal (1), Serious (2), Slight(3)
Independent Variables			Independent Variables		
Light Cond	Nom	Daylight, Lights lit, Darkness	Casualty Class	Nom	Driver, Passenger, Pedestrian
Weather Cond	Nom	Fine, Rain, Snow, Fog	Casualty Age	Cont	Years old
Road Surface	Nom	Dry, Wet, Mud, Snow, Flood,	Casualty Gender	Nom	Male, Female
Driver Gender	Nom	Male, Female	Driver Gender	Nom	Male, Female
Driver Age	Cont	In Years Old	Driver Age	Cont	In Years Old
Vehicle Age	Cont	In Years Old	Vehicle Age	Cont	In Years Old
Vehicle Type	Nom	Bicycle, Motor, Car, Bus, Truck	Vehicle Type	Nom	Bicycle, Motor, Car, Bus, Truck

METHODOLOGY AND ANALYTICS

For both **accident** and **casualty** research questions, given similar nature but different variables, we will utilise several machine learning techniques for **regression** and outcome **classification**, i.e. to classify the severity of accident outcome (fatal, serious or slight) and casualty outcome (fatal, serious or slight) based on our machine learning model. These techniques include but are not limited to **ordinal regression**, **logistic regression**, **KNN**, **decision tree**, **random forest etc** for both research questions.

To ensure robust model selection, steps aimed at improving model accuracy and interpretability will be implemented. This includes **elimination of highly correlated variables** to reduce multicollinearity issues. **Backward selection** techniques will also be applied to identify the most relevant predictors for accident and casualty severity. Additionally, potential interactions between variables to improve the model's predictive performance will be explored.

The dataset will be divided into **training** and **testing** subsets to ensure unbiased performance assessment. To assess the effectiveness of our model, we will use regression metrics including **RSE** and **R-Square (TSS & RSS included)** as well as classification metrics including **precision, recall, accuracy, and the F1-score** whichever is empirically applicable to the resulting model. Furthermore, we will utilize a **confusion matrix** to visualize prediction outcomes, assess misclassification rates, and better understand model behavior across different severity categories.

PRACTICAL IMPLICATIONS

This dataset has significant real-world applications across public safety & business sectors, making it valuable for policymakers, insurance companies, and the automotive industry.

From a **public interest perspective**, the research provides critical insights into accident trends, including the demographics of those affected, common causes, and how different factors contribute to accidents. Since road safety is a major public concern, these insights can help authorities and communities identify high-risk situations and take proactive steps to prevent accidents. Additionally, the dataset highlights the human impact of road incidents, showing how driving patterns and vehicle technologies influence accident outcomes. This information can be instrumental in designing more effective safety campaigns, improving driver education, and advocating for better road infrastructure.

From a **business standpoint**, the dataset offers valuable applications for multiple industries. For insurance companies, both in the car and personal insurance sectors, this data can be used to evaluate accident risks, refine claim processing policies, and adjust premium pricing models based on identified trends. The information on accident severity, casualty demographics, and vehicle conditions provide essential input in predicting the likelihood of claims and managing financial risk. Meanwhile, **car manufacturers and dealerships** can use accident data to improve vehicle safety features to develop safer vehicles and advanced safety technologies to reduce the frequency and impact of crashes.

Additionally, **government and transportation agencies** can leverage these insights to enhance road safety policies, improve infrastructure planning, and develop more targeted campaigns. By understanding how accidents occur and which factors contribute to them, policymakers can create data-driven strategies to improve overall transportation safety.

Overall, the dataset plays a crucial role in supporting informed decision-making across various sectors. Its insights can help minimize accident rates, improve safety regulations, and enhance vehicle design, ultimately contributing to a safer and more efficient transportation system.